

DS701 Homework 4: Probability, MLE, and EM

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Name: _____ BU ID: _____

How this works. Answer **by hand on paper** (or a tablet with a stylus), show your work, then scan and upload to Gradescope, matching each problem to its pages. This is deliberate practice for the closed-notes quizzes, which are in the same style — so do it as much as possible without a computer or an AI assistant. If you are unable to do it without assistance, try AI assistance in “socratic” mode, where you preface your questions by telling AI it is a tutor helping you learn rather than give you answers right away. This homework is **participation-graded**: a serious attempt at every problem earns full credit, and worked solutions are released the day after the deadline. Budget 1–2 hours.

Covers: Lecture 05 (Probabilistic Modeling) and Lecture 06 (Clustering III: GMM and EM). Bayes’ rule for events, odds, and the Bernoulli pmf are stated in the problems. Give numerical answers to 3 decimal places; keep the exact fractions when they are simple.

Problem 1: A screening test, three ways

Practices: Bayes’ rule with the base rate, and the odds / likelihood-ratio form.

A screening test for a disease D has **sensitivity** $P(+ | D) = 0.90$ (positive in 90% of people who have D) and **specificity** $P(- | \bar{D}) = 0.90$ (negative in 90% of people who do not). Bayes’ rule for events, with the denominator expanded by the law of total probability (with $C_k \in \{D, \bar{D}\}$):

$$P(D | +) = \frac{P(+ | D) P(D)}{P(+ | D) P(D) + P(+ | \bar{D}) P(\bar{D})}.$$

- (a) In the general population the **prevalence** (base rate) is $P(D) = 0.01$. A randomly chosen person tests positive. Compute $P(D | +)$, showing the two terms of the denominator separately. In one sentence, why is the answer so far below the 90% that “the test is 90% accurate” suggests? $P(D | +) =$ _____

- (b) The same test is used in a high-risk clinic where $P(D) = 0.10$. Recompute $P(D \mid +)$. Then state the practical lesson in one or two sentences: the test did not change — what did, and what does that say about using one test both for mass screening and in a specialist clinic? $P(D \mid +) = \underline{\hspace{2cm}}$

- (c) **Odds form.** Define the odds of an event as $O(D) = P(D)/P(\bar{D})$ and the positive likelihood ratio as $LR_+ = P(+ \mid D)/P(+ \mid \bar{D})$. Show, starting from Bayes' rule written once for D and once for $\bar{D}$, that

$$O(D \mid +) = LR_+ \times O(D)$$

(posterior odds = likelihood ratio $\times$ prior odds — no denominator needed). Optionally, then compute LR_+ for this test, redo (a) and (b) in odds and convert back to probabilities to confirm your answers, and finally: a person from the general population tests positive on **two independent** administrations of the test — what is $P(D \mid ++)$? (Independence here means the second test multiplies the odds by LR_+ again.) $LR_+ = \underline{\hspace{2cm}}$ $P(D \mid ++)$ =

Problem 2: Maximum likelihood, from scratch

Practices: the MLE recipe — likelihood, log, derivative, solve — on two models, and knowing why the log is there.

- (a) **Bernoulli.** Each observation is $x_n \in \{0, 1\}$ with $P(x_n = 1) = p$, so $P(x_n | p) = p^{x_n}(1-p)^{1-x_n}$. For N i.i.d. observations with $k = \sum_n x_n$ ones: write the likelihood $L(p)$, the log-likelihood $\ell(p)$, set $d\ell/dp = 0$, and solve for $\hat{p}$. Then evaluate $\hat{p}$ for the data 1, 0, 1, 1, 0, 1, 1, 0, 1. (Say in one line why the critical point is a maximum and not a minimum.) For the data, $\hat{p} =$
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- (b) **Gaussian mean and variance.** Start from the Gaussian log-likelihood

$$\ell(\mu, \sigma^2) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2.$$

Derive $\hat{\mu}$ from $\partial\ell/\partial\mu = 0$. Then treat $v = \sigma^2$ as the variable, derive $\hat{\sigma}^2$ from $\partial\ell/\partial v = 0$ (substituting $\hat{\mu}$), and evaluate both on the data 2, 3, 5, 6, 9. Compare $\hat{\sigma}^2$ with the sample variance s^2 that divides by $N - 1$ — which is larger, and by what factor here? $\hat{\mu} =$ _____
 $\hat{\sigma}^2 =$ _____ $s^2 =$ _____ factor: _____

- (c) **Why the log?** Give the two reasons the lecture gives for maximizing ℓ rather than L : one about *where* the maximum is, one about the *form* of the function. Then add the practical one: for $N = 2000$ observations from (a), every factor of L is at most 1 — say roughly how small L is at best (e.g. compare 0.5^{2000} with the smallest positive double, about 10^{-308}) and why ℓ does not have this problem.

Problem 3: One EM iteration by hand

Practices: the E-step and M-step formulas on numbers small enough to see what they do, and the argument for why EM behaves.

A one-dimensional mixture of $K = 2$ Gaussians, both with variance $\sigma^2 = 1$ (**keep the variances fixed at 1 throughout** — update only means and weights). Current parameters: $\mu_1 = 0$, $\mu_2 = 2$, $\pi_1 = \pi_2 = 0.5$. Data: $x_1 = 0$, $x_2 = 1$, $x_3 = 2$, $x_4 = 4$. Useful values: $e^{-0.5} \approx 0.607$, $e^{-2} \approx 0.135$, $e^{-8} \approx 0.00034$; note that with equal variances the factor $1/\sqrt{2\pi}$ is the same in every term and cancels in the responsibilities.

- (a) **E-step.** For each data point compute the responsibilities γ_{i1}, γ_{i2} (3 decimals). Before computing, say for which point the answer must be exactly 0.5 and why. (γ_{i1}, γ_{i2}) for $x_1, \dots, x_4$: _____

- (b) **M-step.** Compute N_1, N_2 , the new means μ'_1, μ'_2 and the new weights π'_1, π'_2 . In one sentence each: why did μ_2 move more than μ_1 , and why did the weights become unequal? $N_1 =$ _____ $N_2 =$ _____ $\mu'_1 =$ _____ $\mu'_2 =$ _____ $\pi'_1 =$ _____ $\pi'_2 =$ _____

- (c) The lecture states that the log-likelihood $\ell(\theta)$ never decreases from one EM iteration to the next. Explain the mechanism in words: what the E-step computes, what the M-step maximizes and why that has a closed form, and why alternating the two cannot lower ℓ . Then say what this guarantees about the sequence $\ell^{(t)}$, what it does **not** guarantee, and what is done about that in practice (relate to Homework 3, Problem 1).
- (d) **Hard vs. soft.** Without recomputing, describe what the four responsibilities in (a) would look like if both variances were $\sigma^2 = 0.01$ instead of 1, and what the M-step for the means would then reduce to. Which algorithm is that? State the one practical difference and the one conceptual difference between GMM/EM and k -means that the lecture draws.

